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I. DIFFUSION MODELS

Diffusion models [1, 2] are a class of generative models that learn a data distribution $p_{\rm data}(\phi)$ by reversing a gradual noising process. The forward process progressively corrupts a data sample $\phi(0) \sim p_{\rm data}$ into pure noise $\phi(T)$ over a time interval $t \in [0, T]$, while the reverse process learns to denoise and thereby generate new samples from the data distribution. Unlike one-shot generators such as GANs or VAEs, diffusion models decompose generation into many small denoising steps, each of which is a tractable learning problem. This decomposition is the source of their strong mode coverage and high sample fidelity.

A. Forward process as a stochastic differential equation

The forward noising process can be formulated as an Itô stochastic differential equation [2]

$$\mathrm{d}\phi \;=\; f(\phi, t)\,\mathrm{d}t \;+\; g(t)\,\mathrm{d}w, \tag{1}$$

where $f(\phi, t)$ is the drift coefficient, $g(t)$ the diffusion coefficient, and w a standard n-dimensional Wiener process. Two canonical choices have emerged.

a. Variance-preserving (VP) SDE. With $f(\phi, t) = -\frac{1}{2}\beta(t)\phi$ and $g(t) = \sqrt{\beta(t)}$, equation (1) reduces to a time-inhomogeneous Ornstein–Uhlenbeck process whose marginal variance is bounded. Denoting $\bar{\alpha}(t) = \exp\!\big(-\int_0^t \beta(s)\,\mathrm{d}s\big)$, the transition kernel is

$$p_{0t}\big(\phi(t)\,\big|\,\phi(0)\big) \;=\; \mathcal{N}\Big(\phi(t);\, \sqrt{\bar{\alpha}(t)}\,\phi(0),\, (1-\bar{\alpha}(t))\,\mathbb{1}\Big). \tag{2}$$

Provided the data are normalised to unit variance, $\mathrm{Var}[\phi(t)]$ is preserved exactly at unity for all t. This is the continuous-time limit of the discrete DDPM formulation of Ho *et al.* [1].

b. Variance-exploding (VE) SDE. With $f(\phi, t) = 0$ and a diffusion coefficient whose square integrates to a prescribed variance schedule $\sigma^2(t)$, one obtains

$$\mathrm{d}\phi \;=\; \sqrt{\frac{\mathrm{d}[\sigma^2(t)]}{\mathrm{d}t}}\,\mathrm{d}w, \tag{3}$$

whose transition kernel is

$$p_{0t}\big(\phi(t)\,\big|\,\phi(0)\big) \;=\; \mathcal{N}\big(\phi(t);\, \phi(0),\, \sigma^2(t)\,\mathbb{1}\big). \tag{4}$$

The signal $\phi(0)$ is preserved exactly and noise of growing variance is added on top. In this convention $\sigma_{\max}$ is typically chosen following the prescription of Song & Ermon [3], $\sigma_{\max} \gtrsim \max_{i,j} \|\phi_i - \phi_j\|_2$, so that $p_{\sigma_{\max}}(\phi)$ covers the entire data manifold. This formulation corresponds to the NCSN/SMLD line of work [4].

Throughout this work we adopt the VE form with the geometric schedule $g(t) = \sigma^t$ on $t \in [0, 1]$ parametrised by a single scalar hyperparameter $\sigma > 1$. The corresponding variance follows by direct integration,

$$\sigma^2(t) \;=\; \int_0^t g(s)^2\,\mathrm{d}s \;=\; \frac{\sigma^{2t} - 1}{2\ln\sigma}, \tag{5}$$

and the endpoint value $\sigma(1) = \sqrt{(\sigma^2 - 1)/(2\ln\sigma)}$ sets the maximum noise scale realised during training. We emphasise that the hyperparameter σ appearing in $g(t) = \sigma^t$ is *not* to be identified with the $\sigma_{\max}$ quantity entering the Song–Ermon prescription [3]: the two are related by (5) evaluated at $t = 1$. Equating them leads to schedules that are drastically miscalibrated for lattice data, where typical field excursions can exceed those of image data by orders of magnitude.

B. Score matching

The key observation enabling reverse-time generation is that the reverse dynamics depend on the data distribution only through the time-dependent *score function*

$$s(\phi, t) \;=\; \nabla_\phi \log p_t(\phi). \tag{6}$$

A neural network $s_\theta(\phi, t)$ is trained to approximate this score via the denoising score-matching objective [5]

$$\mathcal{L}(\theta) \;=\; \mathbb{E}_{t,\,\phi(0),\,\phi(t)|\phi(0)}\Big[\lambda(t)\,\Big\|s_\theta(\phi(t), t) - \nabla_{\phi(t)} \log p_{0t}\big(\phi(t)\,\big|\,\phi(0)\big)\Big\|^2\Big], \tag{7}$$

where $\lambda(t) > 0$ is a weighting function. Because the conditional distribution $p_{0t}(\phi(t) \mid \phi(0))$ is Gaussian with known mean and variance—see (2) and (4)—the target score in (7) admits a closed form, making the objective tractable. In practice this is often reparameterised as predicting the injected noise ϵ (the "ϵ-prediction" of DDPM) or the clean signal $\phi(0)$.

For the VE schedule adopted in this work, the conditional score is elementary,

$$\nabla_{\phi(t)} \log p_{0t}\big(\phi(t)\,\big|\,\phi(0)\big) \;=\; -\frac{\phi(t) - \phi(0)}{\sigma^2(t)} \;=\; -\frac{\epsilon}{\sigma(t)}, \tag{8}$$

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with $\phi(t) = \phi(0) + \sigma(t)\,\epsilon$ and $\epsilon \sim \mathcal{N}(0, \mathbb{1})$. Inserting (8) and choosing $\lambda(t) = \sigma^2(t)$ to equalise the signal-to-noise ratio across the diffusion trajectory yields the practical training objective

$$\mathcal{L}(\theta) \;=\; \mathbb{E}_{t,\,\phi(0),\,\epsilon}\Big[\big\|\sigma(t)\, s_\theta(\phi(t), t) + \epsilon\big\|^2\Big], \qquad (9)$$

which is the loss minimised in all subsequent training runs. Because the VE forward SDE has no drift, its marginals satisfy $p_t \to p_0$ as $t \to 0$, so that a converged network satisfies

$$s_\theta(\phi, t) \;\xrightarrow[t\to 0]{}\; -\nabla_\phi S(\phi). \qquad (10)$$

The small-t limit of the learned score therefore recovers the physical force of the lattice action as a byproduct of training on noise-denoising pairs. The precision of this limit, and the sensitivity of the downstream reverse dynamics to its residual error, underlie the diagnostics of Sec. IV A and Sec. IV B.

C. Reverse-time generation

Given a trained score network, samples are generated by evolving from a tractable prior $\phi(T) \sim \pi$ back to $t = 0$. Two dynamics yield the same marginal distributions $\{p_t\}_{t\in[0,T]}$ [2, 6]. The reverse-time SDE reads

$$\mathrm{d}\phi \;=\; \big[f(\phi, t) - g^2(t)\, s_\theta(\phi, t)\big]\,\mathrm{d}t \;+\; g(t)\,\mathrm{d}\bar{w}, \qquad (11)$$

while the equivalent probability-flow ODE is

$$\frac{\mathrm{d}\phi}{\mathrm{d}t} \;=\; f(\phi, t) - \tfrac{1}{2}\, g^2(t)\, s_\theta(\phi, t). \qquad (12)$$

Although (11) and (12) share the same marginals, they differ substantially at the level of individual trajectories. The SDE is stochastic and non-invertible, but its noise injection acts as a Langevin-like correction that damps accumulated errors from an imperfect score network. The ODE is deterministic and invertible, enabling likelihood evaluation via the continuous change-of-variables formula, latent-space interpolation, and exact inversion. The ODE also admits higher-order numerical solvers (e.g. DPM-Solver [7], the Heun sampler of EDM [8]), allowing high-quality generation in 10–35 function evaluations.

The choice of sampler involves a trade-off. For applications demanding high *spectral fidelity*—for instance when the two-point correlation function $G(p)$ of generated configurations is used as a physically meaningful quality metric, as in the propagator study of Sec. IV—the reverse SDE, or a hybrid stochastic sampler that periodically injects noise into an ODE trajectory, tends to produce more faithful ultraviolet behaviour. A deterministic ODE integrator will faithfully reproduce any bias present in the learned score, whereas the stochasticity of the SDE provides a relaxation mechanism towards the true marginal at each time slice.

II. CRITICAL POINT OF 2D ϕ^4 THEORY FROM WOLFF + FOURIER-ACCELERATED HMC

We consider the scalar ϕ^4 theory on a two-dimensional $L \times L$ square lattice with periodic boundary conditions. The lattice action reads

$$S[\phi] = \sum_x \Big[-2\kappa \sum_{\mu=1}^{2} \phi_x\, \phi_{x+\hat{\mu}} + \phi_x^2 + \lambda\big(\phi_x^2 - 1\big)^2\Big], \quad (13)$$

where κ is the hopping parameter and λ the quartic self-coupling. In two dimensions we set $\lambda = 0.022$. The model undergoes a $\mathbb{Z}_2$-symmetry-breaking phase transition at a critical value $\kappa_c(\lambda)$, which belongs to the 2D Ising universality class.

A. Sampling algorithm

Configurations are generated by a hybrid algorithm that alternates Fourier-accelerated Hybrid Monte Carlo (FA-HMC) updates with Wolff cluster flips.

a. FA-HMC. The standard HMC suffers from critical slowing down because the mass matrix in momentum space does not match the spectrum of the lattice operator. In Fourier-accelerated HMC, one introduces a momentum-dependent mass matrix in Fourier space,

$$M(\hat{p}) = \hat{p}^{\,2} + m_{\mathrm{eff}}^2, \quad \hat{p}^{\,2} = \sum_{\mu=1}^{D} 4\sin^2\!\Big(\frac{\pi k_\mu}{L}\Big), \qquad (14)$$

where m_{eff}^2 is estimated from the variance of Fourier modes during a warm-up phase and subsequently adapted in a series of windows. The conjugate momenta are sampled as $\pi \sim \mathcal{N}(0, M)$ and the kinetic energy is $K = \frac{1}{2}\pi^\top M^{-1}\pi$, both evaluated efficiently via FFT. The leapfrog integrator uses the velocity $v = M^{-1}\pi$ for the position update, ensuring that long-wavelength modes propagate with a step size comparable to that of short-wavelength modes. The step size ε is tuned via dual averaging to maintain an acceptance rate of $\sim 75\%$.

b. Wolff cluster flip. After each FA-HMC trajectory, we perform n_{Wolff} Wolff cluster updates that target the global $\mathbb{Z}_2$ mode. A cluster is grown from a random seed site by adding neighboring site y to the cluster with bond probability

$$P_{\mathrm{bond}}(x, y) = 1 - \exp\!\big(-4\kappa\, \phi_x \phi_y\big), \qquad (15)$$

whenever $\phi_x \phi_y > 0$. All sites in the cluster are then reflected: $\phi_x \to -\phi_x$. This step efficiently decorrelates the magnetization and drastically reduces critical slowing down for the order parameter.

B. Observables

From the spatially averaged magnetization per configuration, $\bar{\phi} = \frac{1}{V}\sum_x \phi_x$ with $V = L^2$, we measure:

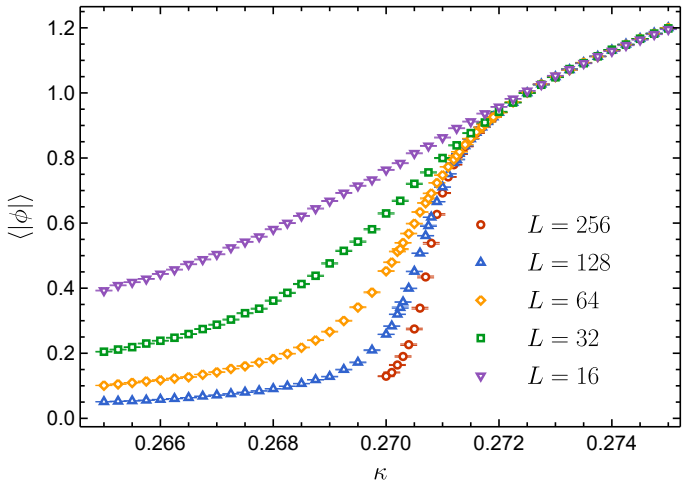

FIG. 1. Order parameter $\langle|\bar{\phi}|\rangle$ as a function of the hopping parameter κ for lattice sizes $L = 16, 32, 64, 128, 256$ at $\lambda = 0.022$. The transition becomes sharper with increasing volume.

1. **Order parameter:** $\langle|\bar{\phi}|\rangle$.

2. **Magnetic susceptibility:** $\chi = V\big(\langle|\bar{\phi}|^2\rangle - \langle|\bar{\phi}|\rangle^2\big)$.

3. **Binder cumulant:** $U_L = 1 - \dfrac{\langle|\bar{\phi}|^4\rangle}{3\,\langle|\bar{\phi}|^2\rangle^2}$.

Statistical errors are estimated by the jackknife method.

C. Results

Fig. 1 shows the order parameter $\langle|\bar{\phi}|\rangle$ as a function of κ for $L = 16, 32, 64, 128, 256$. A clear crossover from the broken phase ($\langle|\bar{\phi}|\rangle \to 0$) to the symmetric phase ($\langle|\bar{\phi}|\rangle > 0$) is observed, with the transition sharpening as L increases.

The magnetic susceptibility χ is shown in Figs. 2 and 3. The peak of χ grows with L and shifts toward the critical κ_c, the expected finite-size scaling $\chi_{\max} \sim L^{\gamma/\nu}$ of the 2D Ising universality class ($\gamma/\nu = 7/4$).

The Binder cumulant U_L is displayed in Fig. 4. In the thermodynamic limit, $U_L \to 2/3$ in the broken phase and $U_L \to 0$ in the symmetric phase. Curves for different L cross at the critical point κ_c, which is a powerful method for locating the phase transition independent of critical exponents.

Fig. 5 shows the Binder crossing analysis. From the intersection of the U_L curves for different lattice-size pairs, we extract the critical hopping parameter

$$\kappa_c(\lambda = 0.022) \approx 0.27088, \tag{16}$$

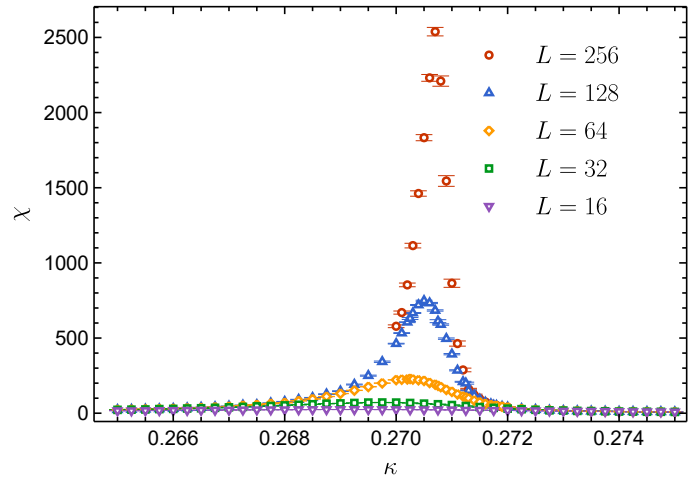

FIG. 2. Magnetic susceptibility χ as a function of κ for different lattice sizes.

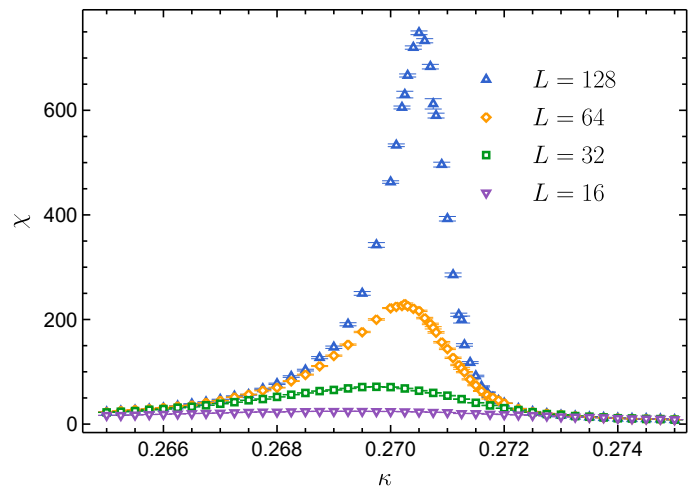

FIG. 3. Magnetic susceptibility χ (alternative representation) as a function of κ.

III. CRITICAL POINT OF 3D ϕ^4 THEORY FROM WOLFF + FA-HMC

We extend the analysis to the three-dimensional ϕ^4 theory on an L^3 cubic lattice. The lattice action takes the same form as Eq. (13) with the sum over nearest neighbors running over $D = 3$ directions. In three dimensions, we set the quartic coupling to $\lambda = 0.9$. The $\mathbb{Z}_2$-breaking transition in 3D belongs to the 3D Ising universality class, with critical exponents $\nu \approx 0.63$.

A. Simulation parameters

B. Results

Fig. 6 shows the order parameter $\langle|\bar{\phi}|\rangle$ as a function of κ for $L = 8, 16, 32, 64$. The crossover from the symmet-

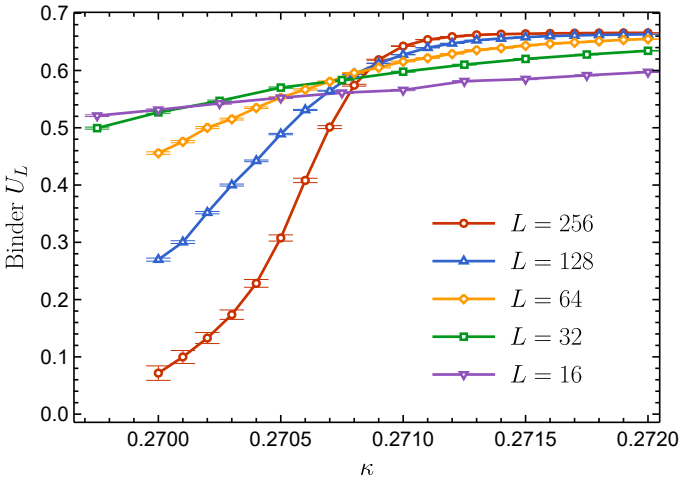

FIG. 4. Binder cumulant U_L as a function of κ for $L = 16, 32, 64, 128, 256$. The crossing of curves at different L determines the critical point κ_c.

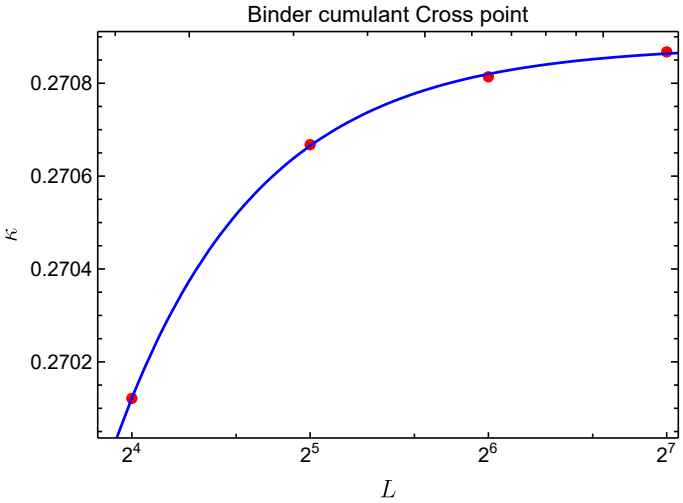

FIG. 5. Binder crossing analysis for the determination of κ_c. The intersection of different lattice-size pairs yields $\kappa_c \approx 0.27088$.

ric to the broken phase sharpens with increasing volume, analogous to the 2D case.

The magnetic susceptibility χ is displayed in Fig. 7. The peak height grows with L and its position converges toward κ_c.

The Binder cumulant U_L is shown in Fig. 8. Curves for different L cross in a narrow region, pinpointing the critical hopping parameter. From the intersection of the $L = 32$ and $L = 64$ curves, we estimate

$$\kappa_c(\lambda = 0.9) \approx 0.19225, \tag{17}$$

consistent with $U_L^* \approx 2/3$ at the crossing, as expected for the 3D Ising universality class.

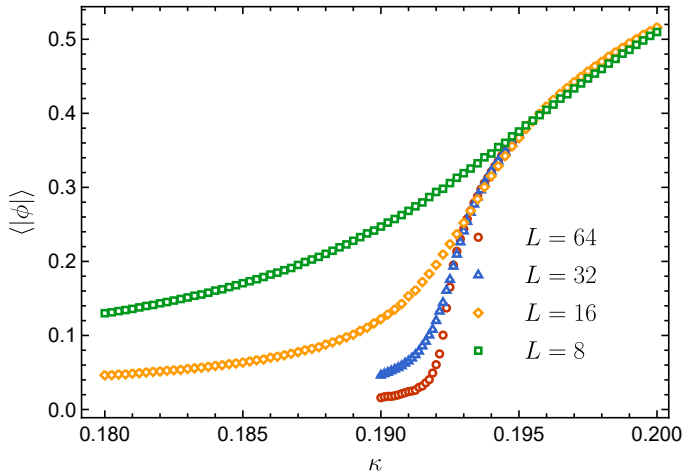

FIG. 6. Order parameter $\langle|\bar{\phi}|\rangle$ vs. κ for the 3D ϕ^4 theory ($\lambda = 0.9$) at lattice sizes $L = 8, 16, 32, 64$.

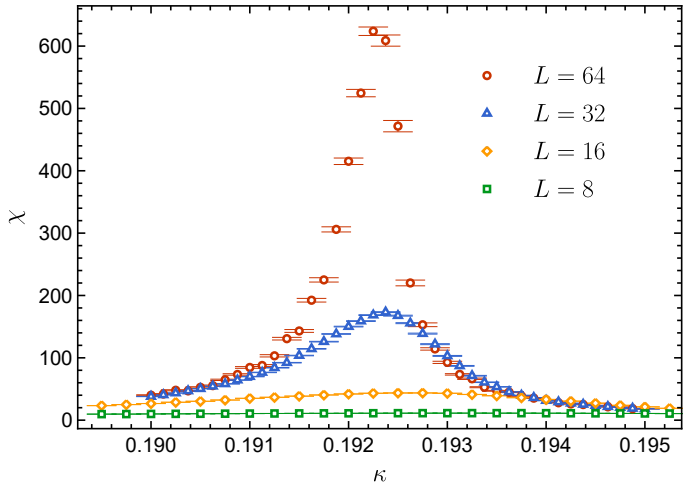

FIG. 7. Magnetic susceptibility χ vs. κ for the 3D theory at different lattice sizes.

IV. MOMENTUM-SPACE PROPAGATOR FROM DIFFUSION MODELS

Having established the critical points for the 2D and 3D ϕ^4 theories via Wolff + FA-HMC, we now test whether a score-based diffusion model can faithfully reproduce the momentum-space propagator $G(p)$. We train a noise-conditional score network (NCSN++) on HMC-generated configurations at $L = 128$ and $\lambda = 0.022$, then generate independent samples by reverse-diffusion and compare the radially-averaged propagator

$$G(|p|) = \langle |\tilde{\phi}(p)|^2 \rangle_{\text{radial}} \tag{18}$$

against the HMC reference, where $\tilde{\phi}(p)$ is the discrete Fourier transform of a configuration. Errors are estimated via bootstrap.

We consider three representative values of the hopping parameter that span the phase diagram:

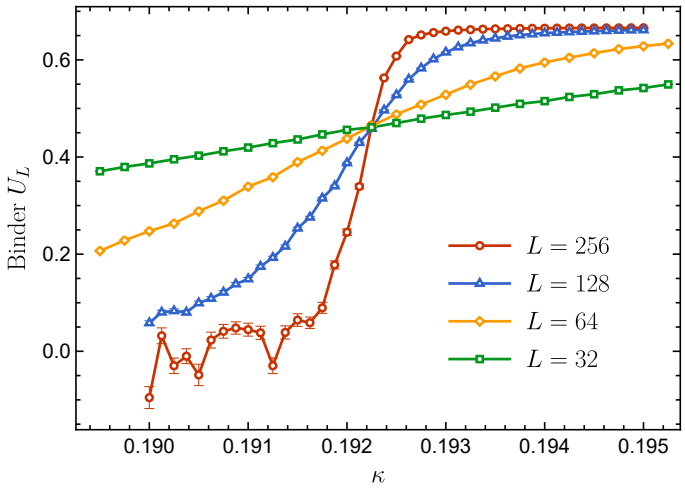

FIG. 8. Binder cumulant U_L vs. κ for the 3D ϕ^4 theory at $L = 8, 16, 32, 64$. The crossing of curves for different L determines $\kappa_c \approx 0.19225$.

- $\kappa = 0.26$ — in the symmetric phase;
- $\kappa = 0.2705 \approx \kappa_c$ — near the critical point;
- $\kappa = 0.28$ — in the broken phase.

A. Acceptance rate as a training diagnostic

A natural quantitative measure of how well a diffusion model has learned the target distribution is the Metropolis–Hastings (MH) acceptance rate of a Metropolis-Adjusted Langevin Algorithm (MALA) [9] that uses the score network as its proposal drift.

In the reverse-diffusion framework, the score network $s_\theta(\phi, t) \approx \nabla_\phi \log p_t(\phi)$ is trained to approximate the score of the noised distribution p_t. At small diffusion time $t \to 0$, $p_t \to p_0 \propto e^{-S}$ and therefore $s_\theta(\phi, t) \to -\nabla_\phi S(\phi)$. From the current state ϕ_τ, the MALA proposal is

$$\psi_{\tau+1} = \phi_\tau + h\, s_\theta(\phi_\tau,\, t_{\rm mh}) + \sqrt{2h}\,\eta, \tag{19}$$

where h is the step size, $t_{\rm mh}$ the diffusion time at which the score is evaluated, and $\eta \sim \mathcal{N}(0, \mathbb{1})$. Conditioned on ϕ_τ, the proposal is a Gaussian with mean $\phi_\tau + h\, s_\theta(\phi_\tau, t_{\rm mh})$ and variance $2h\, I_n$, so the proposal kernel has the explicit form

$$q(\psi_{\tau+1}|\phi_\tau) = \frac{1}{(4\pi h)^{n/2}} \exp\Big(-\frac{1}{4h}\big\|\psi_{\tau+1} - \phi_\tau - h\, s_\theta(\phi_\tau, t_{\rm mh})\big\|^2\Big), \tag{20}$$

where $n = L^D$ is the number of degrees of freedom and $p(\phi) \propto e^{-S(\phi)}$ is the known physical distribution. The accept-reject step is

$$\phi_{\tau+1} = \begin{cases} \psi_{\tau+1} & \text{with prob. } \min\big\{1,\ \frac{p(\psi_{\tau+1})\, q(\phi_\tau|\psi_{\tau+1})}{p(\phi_\tau)\, q(\psi_{\tau+1}|\phi_\tau)}\big\}, \\ \phi_\tau & \text{otherwise.} \end{cases} \tag{21}$$

The transition probability ratio can be computed explicitly as

$$\frac{p(\psi_{\tau+1})\, q(\phi_\tau|\psi_{\tau+1})}{p(\phi_\tau)\, q(\psi_{\tau+1}|\phi_\tau)} = \exp\Big(-S(\psi_{\tau+1}) - \tfrac{1}{4h}\big\|\phi_\tau - \psi_{\tau+1} - h\, s_\theta(\psi_{\tau+1}, t_{\rm mh})\big\|^2 + S(\phi_\tau) + \tfrac{1}{4h}\big\|\psi_{\tau+1} - \phi_\tau - h\, s_\theta(\phi_\tau, t_{\rm mh})\big\|^2\Big). \tag{22}$$

Substituting the proposal (19), i.e. $\psi_{\tau+1} - \phi_\tau - h\, s_\theta(\phi_\tau) = \sqrt{2h}\,\eta$, the log acceptance probability simplifies to

$$\log \alpha = \big[S(\phi_\tau) - S(\psi_{\tau+1})\big] + \tfrac{1}{2}\|\eta\|^2 - \tfrac{1}{4h}\big\|\sqrt{2h}\,\eta + h\big[s_\theta(\phi_\tau, t_{\rm mh}) + s_\theta(\psi_{\tau+1}, t_{\rm mh})\big]\big\|^2, \tag{23}$$

which is the form used in practice. The MH step guarantees that, regardless of the quality of s_θ, the chain targets the correct distribution e^{-S}. The acceptance rate $\langle \min(1, e^{\log\alpha})\rangle$ therefore measures how closely the score-network proposal matches the true Langevin drift $-\nabla_\phi S$: a well-trained model yields a high acceptance rate, while

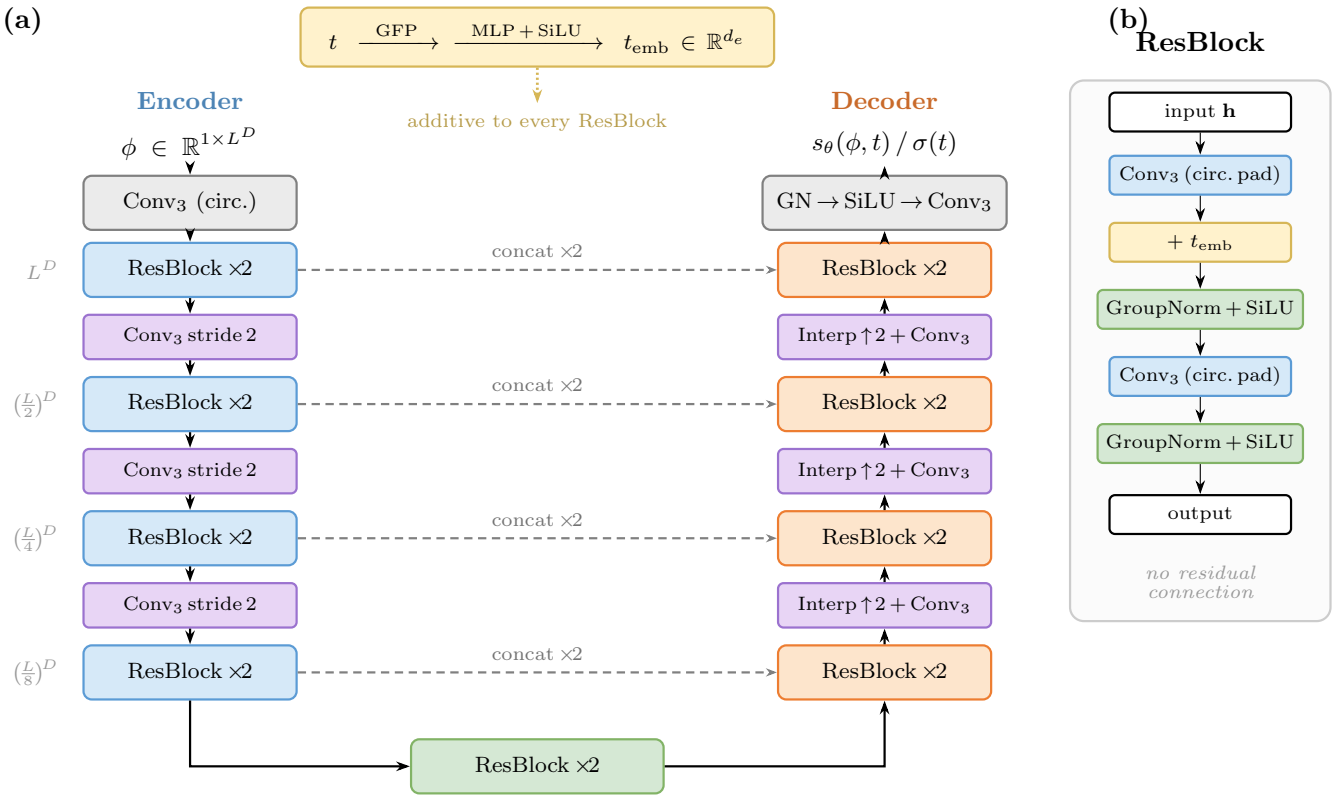

FIG. 9. Architecture of the NCSN++ score network used for both 2D and 3D ϕ^4 lattice field theory. **(a)** U-Net overview. The encoder (blue) consists of four ResBlock pairs at progressively coarser resolutions, separated by stride-2 convolutions (purple) that halve the spatial dimensions. The decoder (orange) mirrors this structure, upsampling via nearest-neighbor interpolation followed by a convolution (purple). Each encoder level provides two skip connections—from both sub-blocks—to the corresponding decoder level via channel concatenation ("concat ×2"). The diffusion time t is encoded by Gaussian Fourier projection (GFP) and a learned MLP, then injected additively into every ResBlock. The final output is divided by $\sigma(t)$. **(b)** Internal structure of a single ResBlock; no residual connection is used. All convolutions use circular padding to enforce periodic boundary conditions. For 2D fields all convolutions are `Conv2d`; for 3D they become `Conv3d` ($D = 3$).

a poorly trained one is rejected more often.

a. Parameter calibration. Two parameters must be chosen: the step size h and the diffusion time t_{mh}. The calibration criterion is the *diagnostic power*, defined as the difference in acceptance rate between a late-epoch (well-trained) and an early-epoch (poorly trained) checkpoint.

b. Step-size scaling. For MALA on a d-dimensional i.i.d. product target $\pi(x) = \prod_i f(x_i)$, Roberts and Rosenthal [9] proved that the optimal proposal variance scales as $h \sim d^{-1/3}$, yielding an optimal acceptance rate of 0.574. For a D-dimensional lattice with $d = L^D$ sites, this would give $h \sim L^{-D/3}$, i.e. $h \sim L^{-2/3}$ in 2D and $h \sim L^{-1}$ in 3D. However, the ϕ^4 action contains nearest-neighbour couplings ($-2\kappa \sum_{\langle x,y\rangle} \phi_x \phi_y$) that strongly violate the i.i.d. assumption. The total action change for a global proposal scales as $\Delta S \sim d \cdot h$, so maintaining $\Delta S \sim O(1)$ for reasonable acceptance requires the RWM-type scaling $h \sim 1/d = 1/L^D$ rather than the MALA-optimal $d^{-1/3}$. Empirically, $h \sim L^{-D/3}$ produces a step size that is far too large in both 2D and 3D. We therefore adopt $h = c/L^D$ throughout, i.e. $h = c/L^2$ in 2D and $h = c/L^3$ in 3D, with the coefficient c calibrated independently for each dimensionality.

We find that the diagnostic power is maximised at $t_{\text{mh}} = 10^{-4}$, which is much smaller than values typically used in sampling ($t \sim 10^{-2}$). As shown in Fig. 11, at large $t_{\text{mh}} \gtrsim 5 \times 10^{-3}$ the gap reverses sign: the score at large t approximates the score of a heavily smeared distribution $p_t \neq p_0$, so a well-trained model's proposal is rejected more often by the exact action than a poorly trained one whose drift is weak and effectively random. This reversal makes large t_{mh} unsuitable as a diagnostic. For the step size in 2D, Fig. 10 shows that $c = 0.2$ (i.e. $h = 0.2/L^2$) maximises the diagnostic power while keeping the late-epoch acceptance rate in the informative range 50–80%.

With the calibrated parameters $c = 0.2$ and $t_{\text{mh}} = 10^{-4}$, we measure the MALA acceptance rate using $N = 1024$ reference configurations drawn from the training data and a single MH step per configuration. Using a single MH step avoids the warm-up bias that arises

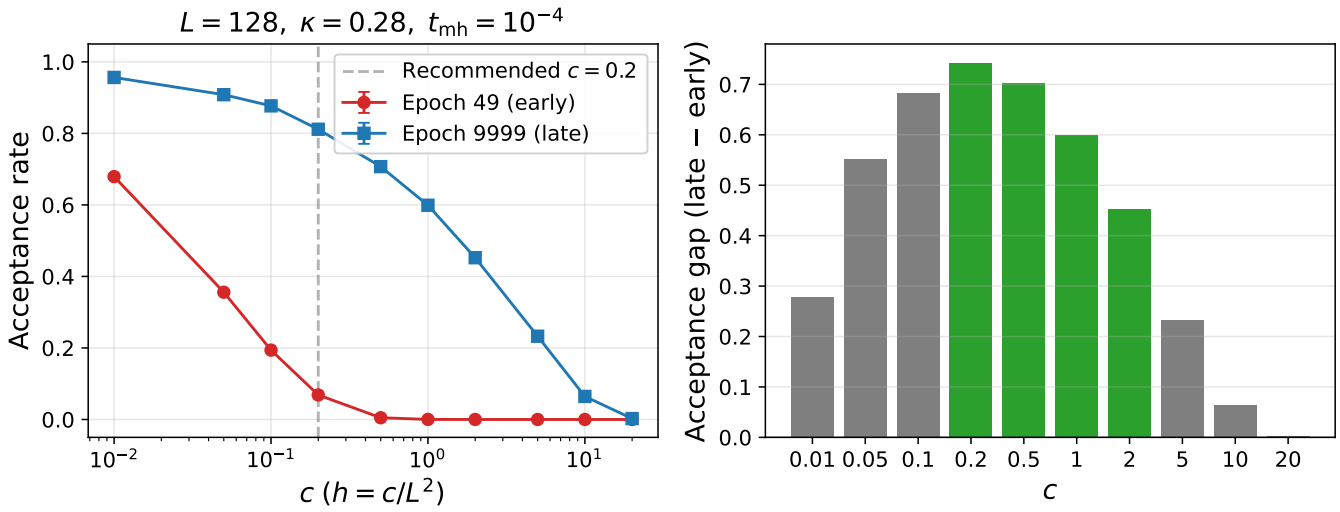

FIG. 10. Calibration of the MALA step size $h = c/L^2$ at $L = 128$, $\kappa = 0.28$, with $t_{\rm mh} = 10^{-4}$, $N = 1024$ reference configurations and a single MH step. Left: acceptance rate vs. c for the early-epoch (epoch 49, poorly trained) and late-epoch (epoch 9999, well-trained) checkpoints. Right: diagnostic power (acceptance gap between late and early checkpoints). Green bars indicate the "useful" range where the late-epoch acceptance rate is between 30% and 85%.

FIG. 11. Calibration of the evaluation time $t_{\rm mh}$ at $L = 128$, $\kappa = 0.28$, with $c = 0.2$, $N = 1024$ and a single MH step. Left: acceptance rate vs. $t_{\rm mh}$ for early (epoch 49) and late (epoch 9999) checkpoints. Right: diagnostic power. The gap reverses sign for $t_{\rm mh} \gtrsim 5 \times 10^{-3}$, where the score network evaluates the gradient of a heavily smeared distribution rather than the physical action.

with multiple steps (where the MH chain migrates toward lower-action regions, artificially inflating the acceptance rate) and maximises statistical independence across samples.

c. Acceptance rate during training. Fig. 12 shows the acceptance rate as a function of training epoch for all three κ values at $L = 128$. In all three cases, the acceptance rate starts near zero for the untrained model (epoch 49) and rises rapidly during the first ∼500 epochs, eventually fluctuating in the range 60–90%. The substantial epoch-to-epoch fluctuations reflect the stochastic nature of both the score-network training and the finite-sample MALA estimator, but the overall upward trend provides a clear signal that the score network progressively learns the correct Langevin drift.

d. Acceptance rate across lattice sizes. Tabs. I and II summarise the average MALA acceptance rate for all trained models at two stages of training: an early window (epochs 49–249) and a late window (epochs 3999–4999). In all cases, the late-epoch acceptance rate is substantially higher than the early-epoch value, confirming that the diagnostic tracks training progress. The late-epoch acceptance rate increases mildly with lattice size, from $\sim$ 63–71% at $L = 8$ to $\sim$ 75–82% at $L = 64$–128. This trend is consistent with the $h = c/L^2$ scaling being slightly more conservative than the true optimal scaling for the correlated ϕ^4 target: the optimal exponent likely lies between $L^{-2/3}$ (i.i.d. limit [9]) and L^{-2} (our choice), so at larger L the step size is increasingly conservative and the acceptance rate rises accordingly. Despite this mild L-dependence, the acceptance rate remains a useful diagnostic at all lattice sizes tested.

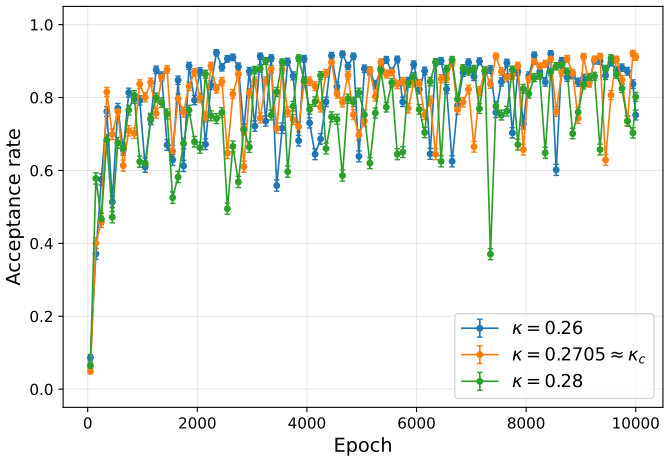

FIG. 12. MALA acceptance rate vs. training epoch for the 2D ϕ^4 model at $L = 128$, with $c = 0.2$, $t_{\rm mh} = 10^{-4}$, $N = 1024$ and a single MH step, for three values of κ spanning the phase diagram.

TABLE I. Average MALA acceptance rate over epochs 49–249 (early training), with $c = 0.2$, $t_{\rm mh} = 10^{-4}$, $N = 1024$ and a single MH step.

L	$\kappa = 0.26$	$\kappa = 0.2705$	$\kappa = 0.28$
8	0.152	0.136	0.129
16	0.082	0.228	0.031
32	0.207	0.205	0.297
64	0.288	0.375	0.300
128	0.336	0.325	0.296

TABLE II. Same as Tab. I but averaged over epochs 3999–4999 (late training).

L	$\kappa = 0.26$	$\kappa = 0.2705$	$\kappa = 0.28$
8	0.635	0.715	0.627
16	0.683	0.641	0.693
32	0.680	0.753	0.683
64	0.827	0.818	0.747
128	0.790	0.792	0.749

B. Direct score quality diagnostic

While the MALA acceptance rate provides an effective indirect measure of score quality, it depends on the step size h and evaluation time $t_{\rm mh}$, requiring careful calibration (Fig. 10, Fig. 11). A more direct diagnostic is to compare the model output $s_\theta(\phi, t)$ against the true gradient $-\nabla_\phi S(\phi)$ at a small diffusion time t, without any Metropolis–Hastings sampling.

For each reference configuration ϕ drawn from the training data, we evaluate the model score $s_\theta(\phi, t_{\rm eval})$ at $t_{\rm eval} = 10^{-4}$ and compute the cosine similarity

$$C = \frac{\langle s_\theta, -\nabla S\rangle}{\|s_\theta\|\ \|\nabla S\|}, \tag{24}$$

the relative mean-squared error $\|s_\theta - (-\nabla S)\|^2/\|\nabla S\|^2$, and the magnitude ratio $\|s_\theta\|/\|\nabla S\|$, averaged over $N = 512$ configurations. The choice $t_{\rm eval} = 10^{-4}$ (noise level $\sigma \approx 0.01$, or 0.5% of the data range in normalised space) ensures that $p_t \approx p_0$ to high accuracy while the score estimate remains numerically stable. A larger evaluation time $t = 0.01$ ($\sigma \approx 0.10$) introduces non-negligible Gaussian smoothing that systematically reduces the magnitude ratio below unity even for a perfectly trained model, making it a less informative diagnostic; a smaller time $t = 10^{-5}$ ($\sigma \approx 0.003$) enters the regime where division by $\sigma(t)$ in the network output amplifies numerical noise.

Fig. 13 shows all three metrics as a function of training epoch for $L = 128$ at all three κ values. The cosine similarity rises from ~ 0.99 at epoch 49 to ~ 0.998 at late training; the relative MSE drops by more than an order of magnitude; and the magnitude ratio converges toward unity from above, starting at ~ 1.5 and reaching ~ 0.9–1.0 by epoch 5000. These trends are consistent with the monotonic improvement observed in the MALA acceptance rate (Fig. 12) and confirm that the score network progressively learns both the direction and the magnitude of $-\nabla S$.

TABLE III. Average cosine similarity C and magnitude ratio $\|s_\theta\|/\|\nabla S\|$ over epochs 49–249 (early training), evaluated at $t = 10^{-4}$ with $N = 512$ reference configurations.

	Cosine similarity C			Magnitude ratio		
L	κ=0.26	0.2705	0.28	κ=0.26	0.2705	0.28
8	0.987	0.971	0.974	2.102	2.018	1.942
16	0.985	0.977	0.976	2.046	1.716	2.042
32	0.981	0.981	0.986	1.704	1.647	1.568
64	0.992	0.985	0.990	1.581	1.401	1.408
128	0.994	0.994	0.989	1.512	1.472	1.459

TABLE IV. Same as Tab. III but averaged over epochs 3999–4999 (late training).

	Cosine similarity C			Magnitude ratio		
L	κ=0.26	0.2705	0.28	κ=0.26	0.2705	0.28
8	0.988	0.991	0.990	1.282	1.148	1.267
16	0.989	0.989	0.993	1.119	1.158	1.052
32	0.988	0.993	0.990	1.053	1.014	0.953
64	0.996	0.996	0.996	0.968	0.972	0.949
128	0.997	0.997	0.998	0.907	0.938	0.950

Tabs. III and IV summarise the average cosine similarity and magnitude ratio across all lattice sizes and κ values, in the same early and late training windows used for the acceptance rate (Tabs. I and II). In the early window (epochs 49–249), the magnitude ratio is ~ 1.4–2.1, indicating that the untrained model significantly overshoots the true gradient magnitude. By late training (epochs 3999–4999), the magnitude ratio converges to 0.9–1.1 for $L \geq 32$, confirming quantitative accuracy of the learned score. For $L = 8$ and 16 a slight residual overshoot (~ 1.1–1.3) persists, likely reflecting the smaller number of degrees of freedom where the model has less room to average out local fluctuations.

C. Training loss for 3D ϕ^4

We train the NCSN++ model on Wolff + FA-HMC configurations of the 3D theory at $L = 64$ and $\lambda = 0.9$ for three hopping parameters: $\kappa = 0.18$ (symmetric phase), $\kappa = 0.1923 \approx \kappa_c$ (near the critical point), and $\kappa = 0.2$ (broken phase). The score-matching loss is decomposed by the diffusion time t into a UV component ($t < 0.2$, probing short-distance modes), an intermediate component ($0.2 < t < 0.8$), and an IR component ($t > 0.8$, probing long-distance modes).

Fig. 14 shows the training loss evolution up to 20000 epochs. The total loss (panel a) converges for all three κ values, with a clear hierarchy: the symmetric-phase model ($\kappa = 0.18$) has the highest loss plateau, while the broken-phase model ($\kappa = 0.2$) converges to the lowest

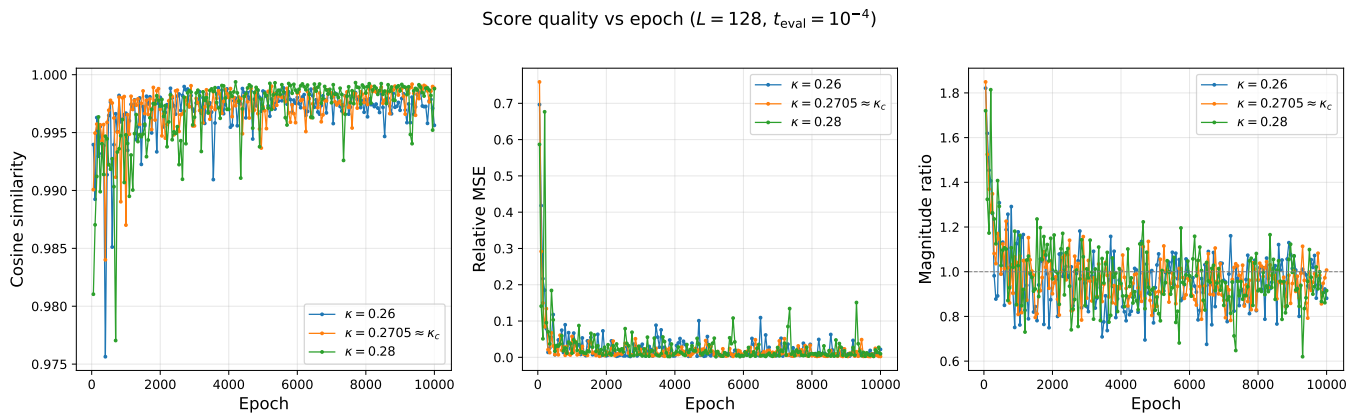

FIG. 13. Score quality vs. training epoch for the 2D ϕ^4 model at $L = 128$, evaluated at $t = 10^{-4}$ with $N = 512$ reference configurations. Left: cosine similarity. Centre: relative MSE. Right: magnitude ratio (dashed line indicates the ideal value of 1).

value. The decomposition reveals that the IR loss (panel d) is the dominant contribution and the most sensitive to κ: the symmetric phase shows the largest IR loss, reflecting the challenge of learning long-wavelength modes when the correlation length is large but finite. The mid and UV components (panels b, c) show a similar ordering, confirming that the phase-dependent difficulty extends across all noise levels.

D. Acceptance rate for 3D ϕ^4

We apply the same MALA acceptance-rate diagnostic to the 3D models at $L = 64$, $\lambda = 0.9$, now with $h = c/L^3$ and $N = 256$ reference configurations. Fig. 15 shows the step-size calibration: the optimal coefficient is again $c = 0.2$, the same value found in 2D. This consistency strongly supports the $h = c/L^D$ scaling: the coefficient c is universal across spatial dimensions once the volume factor L^D is correctly accounted for. The $t_{\rm mh}$ calibration (Fig. 16) with $c = 0.5$ confirms that $t_{\rm mh} = 10^{-4}$ maximises the diagnostic power, as in the 2D case.

Fig. 17 shows the acceptance rate vs. training epoch for all three 3D models with $c = 0.5$. The acceptance rates reach 60–90% at late epochs, comparable to the 2D results and confirming that the $h = c/L^D$ scaling yields a consistent diagnostic across dimensions.

a. Direct score quality for 3D. We apply the same direct score-quality diagnostic to the 3D models, evaluating the cosine similarity, relative MSE, and magnitude ratio at $t_{\rm eval} = 10^{-4}$ with $N = 64$ reference configurations. Fig. 18 shows the three metrics as a function of training epoch for all three κ values. The cosine similarity rises from ~ 0.96 at early training to ~ 0.995 at late training; the relative MSE drops by more than an order of magnitude; and the magnitude ratio converges to ~ 0.97–1.00 from an initial overshoot of ~ 1.4–1.5. These trends mirror the 2D results (Fig. 13) and confirm that the score network learns both the direction and the magnitude of $-\nabla S$ in three dimensions.

E. Quantitative propagator-mismatch metrics

To quantify how well the diffusion-model propagator $G_{\rm DM}(k)$ matches the reference $G_{\rm train}(k)$ across momentum bins, one could compare them via absolute deviation $\Delta G(k) = G_{\rm DM} - G_{\rm train}$ or relative deviation $\Delta G/G_{\rm train}$, but neither scales uniformly across the UV/IR range. The absolute deviation is dominated by a handful of low-k bins where G is large, while the relative deviation grows artificially at high k where G is tiny and the mismatch contributes negligibly to the field variance. We instead use three complementary metrics, each with a sharp probabilistic or statistical interpretation.

a. (i) Per-mode Gaussian KL. Under translation invariance, any zero-mean Gaussian field measure diagonalises in momentum with covariance $G(k)$, and the modes become independent. The Kullback–Leibler divergence between the two Gaussian field distributions $p_{\rm train} = \mathcal{N}(0, G_{\rm train})$ and $p_{\rm DM} = \mathcal{N}(0, G_{\rm DM})$ therefore reduces to an additive sum over momentum modes:

$$D_{\rm KL}(p_{\rm train}\|p_{\rm DM}) = \sum_k D_k, \qquad D_k = \tfrac{1}{2}\big[r_k - 1 - \log r_k\big], \quad r_k = \frac{G_{\rm train}}{G_{\rm DM}(} \tag{25}$$

The function $f(r) = r - 1 - \log r$ vanishes at $r = 1$, is strictly convex, grows linearly as $r \to \infty$ (linear penalty for DM underestimating the variance) and only logarithmically as $r \to 0^+$ (weaker penalty for overestimation). For small relative deviations $r = 1+\varepsilon$ one has $D_k \simeq \varepsilon^2/4$, so D_k reduces to the usual quadratic relative-deviation measure in the well-trained regime, but is far tamer than $(r-1)^2$ for large mismatches in the UV tail. Because D_k is dimensionless and additive, contributions from different $|k|$-bins—and hence UV vs IR halves—can be compared and summed on equal footing.

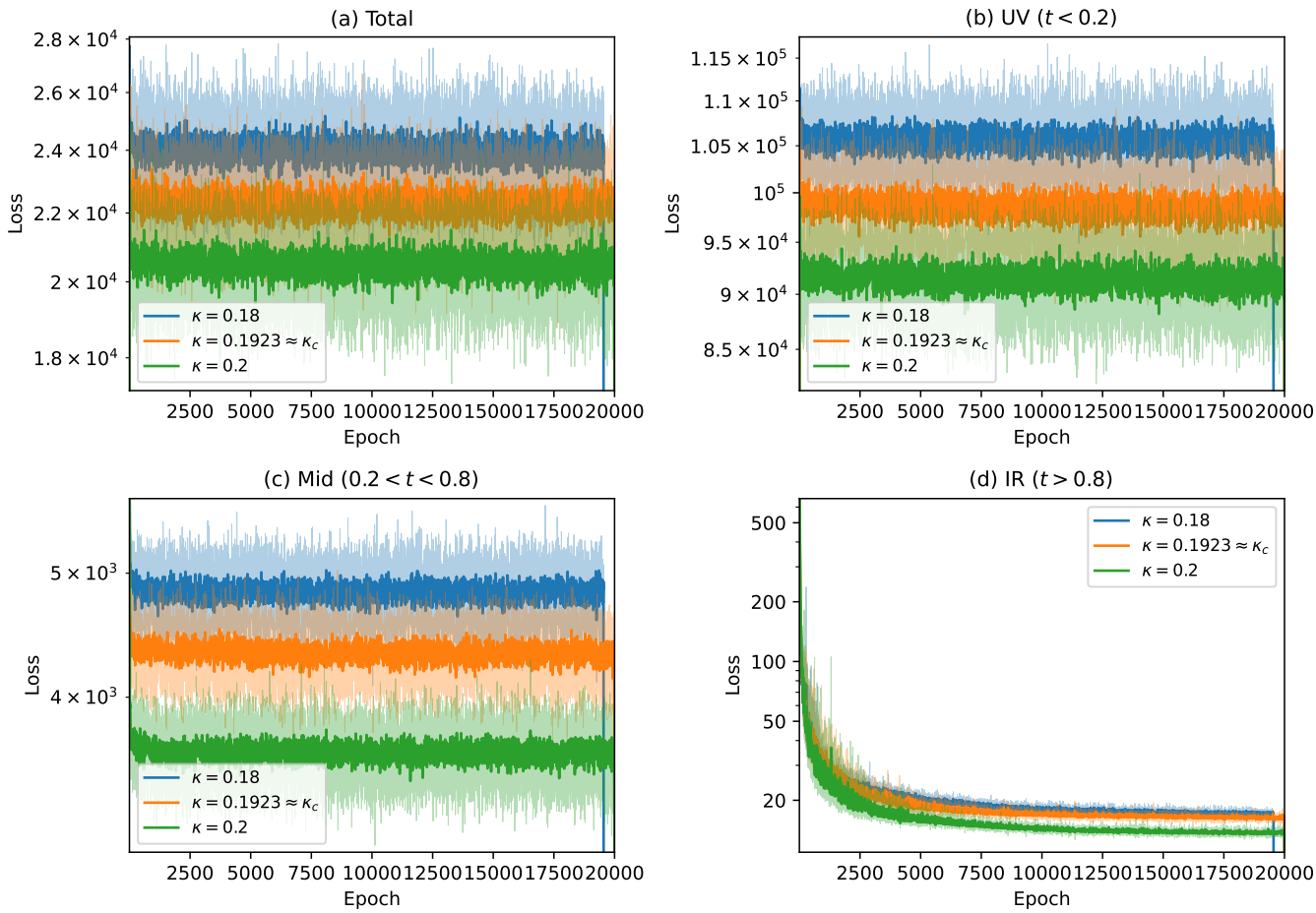

FIG. 14. Training loss vs. epoch for the 3D ϕ^4 NCSN++ diffusion model at $L = 64$, $\lambda = 0.9$: (a) total loss, (b) UV component ($t < 0.2$), (c) mid component ($0.2 < t < 0.8$), (d) IR component ($t > 0.8$). Each panel compares three κ values spanning the phase diagram.

Crucially, (25) is the KL between two *Gaussian surrogates* determined by the two-point functions alone. When the underlying field theory is non-Gaussian (as in ϕ^4 with interactions), the metric captures the second-moment mismatch but is blind to higher-cumulant structure such as the connected four-point function $\langle\phi^4\rangle_{\rm conn}$; matching the two-point function is a necessary but not sufficient condition for distributional equivalence. The generalisation $D_{\rm KL}(p\|q_{\rm Gauss}) = D_{\rm KL}(p_{\rm Gauss}\|q_{\rm Gauss}) + J(p)$, with $J(p) \geq 0$ the negentropy of the true distribution, makes the missing piece explicit.

b. (ii) Statistical z-score. Bootstrap-resampling the configurations yields uncertainty estimates $\sigma_{\rm DM}(k)$ and $\sigma_{\rm train}(k)$ on the two propagators. The standardised residual

$$z_k = \frac{G_{\rm DM}(k) - G_{\rm train}(k)}{\sqrt{\sigma^2_{\rm DM}(k) + \sigma^2_{\rm train}(k)}} \tag{26}$$

tests whether the per-bin discrepancy is statistically significant given the finite sample size. Bins with $|z_k| \lesssim 2$ are consistent with the reference at the 2σ level; persistent deviations with $|z_k| \gg 2$ across many bins signal a real bias rather than sampling noise. The root-mean-square $z_{\rm rms} = \sqrt{\langle z_k^2\rangle}$ provides a single scalar that saturates at $\mathcal{O}(1)$ when the mismatch is at the sample-noise floor and grows without bound when the bias dominates.

c. (iii) Phase-space weighted propagator difference. Expressed in the continuum limit, the field variance decomposes as $\langle\phi^2\rangle = \int d^Dk/(2\pi)^D\, G(k) \propto \int_0^\infty dk\, k^{D-1} G(k)$, so that each radial momentum shell contributes its propagator weighted by the phase-space measure k^{D-1}. Accordingly, the density

$$w_k = k^{D-1}\big[G_{\rm DM}(k) - G_{\rm train}(k)\big] \tag{27}$$

measures, up to an overall $(2\pi)^{-D}$ factor, how much each momentum shell contributes to the total mismatch in $\langle\phi^2\rangle$. Integrating w_k over a UV or IR band directly quantifies the corresponding contribution to the two-point mismatch in position space, with physical units. Compared to D_k and z_k, w_k has a concrete physical interpretation but is not dimensionless, so the three metrics should be viewed as complementary rather than redundant.

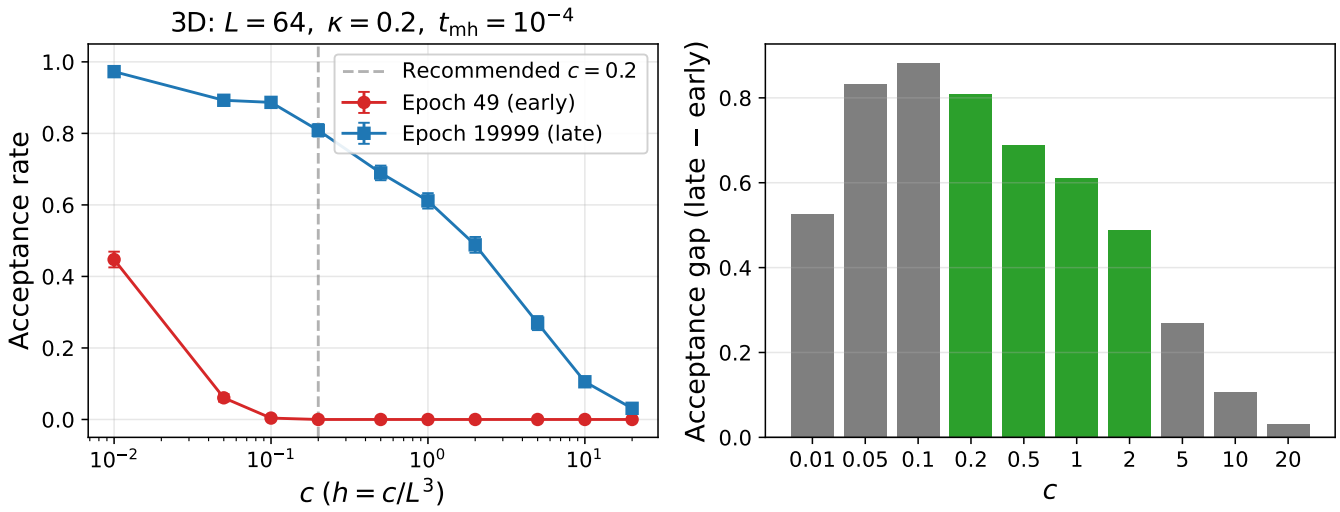

FIG. 15. Calibration of the MALA step size $h = c/L^3$ for the 3D ϕ^4 model at $L = 64$, $\kappa = 0.2$, $\lambda = 0.9$, with $t_{\rm mh} = 10^{-4}$, $N = 256$ and a single MH step. The optimal $c = 0.2$ matches the 2D result, validating the $h = c/L^D$ scaling.

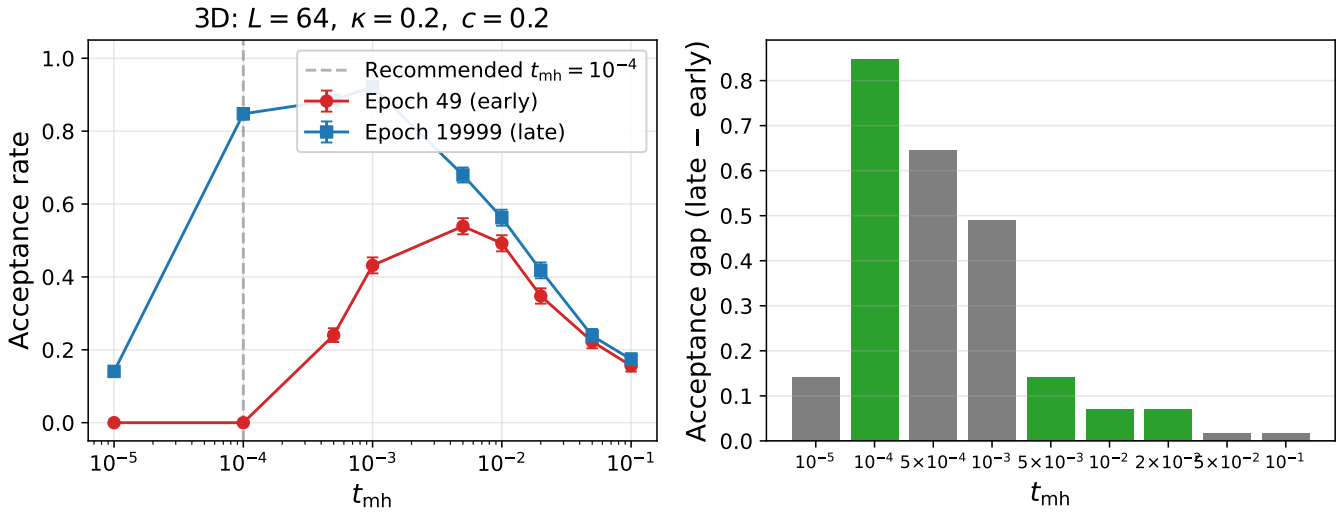

FIG. 16. Calibration of $t_{\rm mh}$ for the 3D model at $L = 64$, $\kappa = 0.2$, with $c = 0.5$ and $h = c/L^3$, $N = 256$ and a single MH step.

d. Illustration on CelebA 64×64. Fig. 19 applies these three diagnostics to a NCSN++ diffusion model trained on CelebA grayscale 64×64 images, using log-spaced checkpoints from epoch 1 to epoch 10,000. Each checkpoint supplies $N = 512$ EM-sampled images; the reference $G_{\rm train}$ is computed from 50,000 training images, so $\sigma_{\rm train} \ll \sigma_{\rm DM}$ and the DM sample size dominates the bootstrap uncertainty. All three metrics display the expected monotonic improvement during training: the per-mode KL drops by nearly two orders of magnitude at all $|k|$, the z-score envelope contracts toward the $\pm 2\sigma$ band, and the phase-space weighted difference w_k shrinks and changes sign as the model passes from underestimating to slightly overestimating the low-k power. The UV (large-$|k|$) portion dominates the total KL throughout training, consistent with the difficulty of learning high-frequency image modes.

F. $\kappa = 0.26$: symmetric phase

Fig. 20 shows the propagator for $\kappa = 0.26$ at several training epochs compared to the HMC baseline. The diffusion model converges to the HMC propagator as training progresses, with the largest deviations observed at early epochs and at small momenta (large correlation lengths).

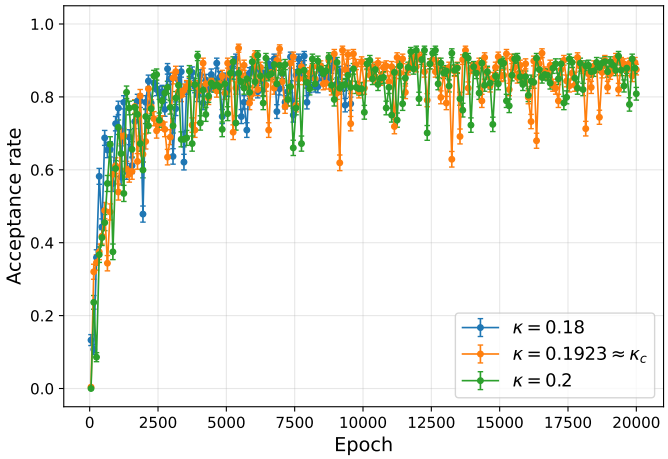

FIG. 17. MALA acceptance rate vs. training epoch for the 3D ϕ^4 model at $L = 64$, $\lambda = 0.9$, with $c = 0.5$, $h = c/L^3$, $t_{\rm mh} = 10^{-4}$, $N = 256$ and a single MH step.

G. $\kappa = 0.2705$: near-critical regime

Near the critical point $\kappa \approx \kappa_c$, the correlation length diverges and the propagator develops a power-law tail $G(p) \sim p^{-(2-\eta)}$ at small momenta, making it the most demanding test case. Fig. 21 demonstrates that the diffusion model successfully captures this behavior, with the DM propagator approaching the HMC result upon sufficient training.

H. $\kappa = 0.28$: broken phase

In the ordered phase at $\kappa = 0.28$, the propagator acquires a large zero-mode contribution and a mass gap. Fig. 22 shows that the diffusion model also reproduces the propagator faithfully in this regime.

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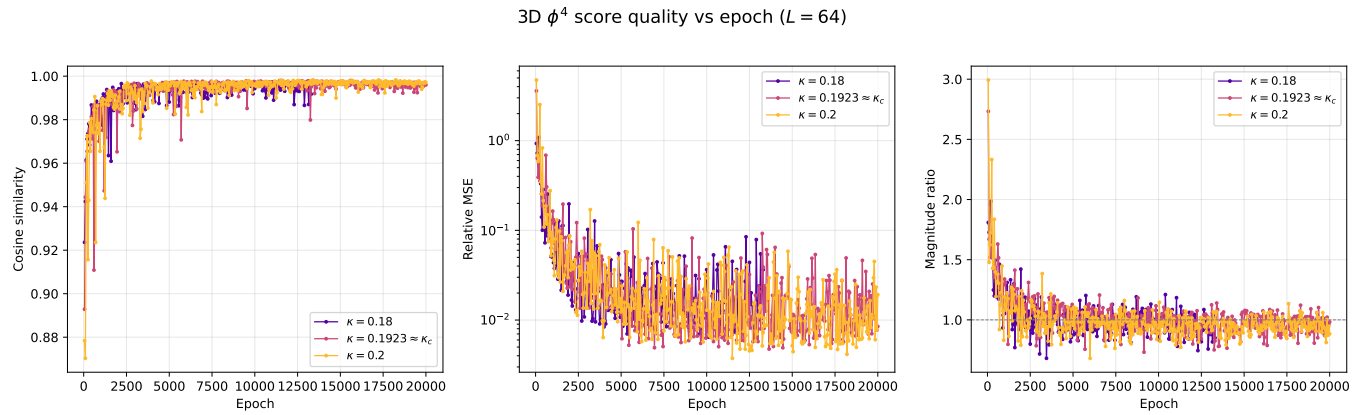

FIG. 18. Score quality vs. training epoch for the 3D ϕ^4 model at $L = 64$, $\lambda = 0.9$, evaluated at $t = 10^{-4}$ with $N = 64$ reference configurations. Left: cosine similarity. Centre: relative MSE. Right: magnitude ratio.

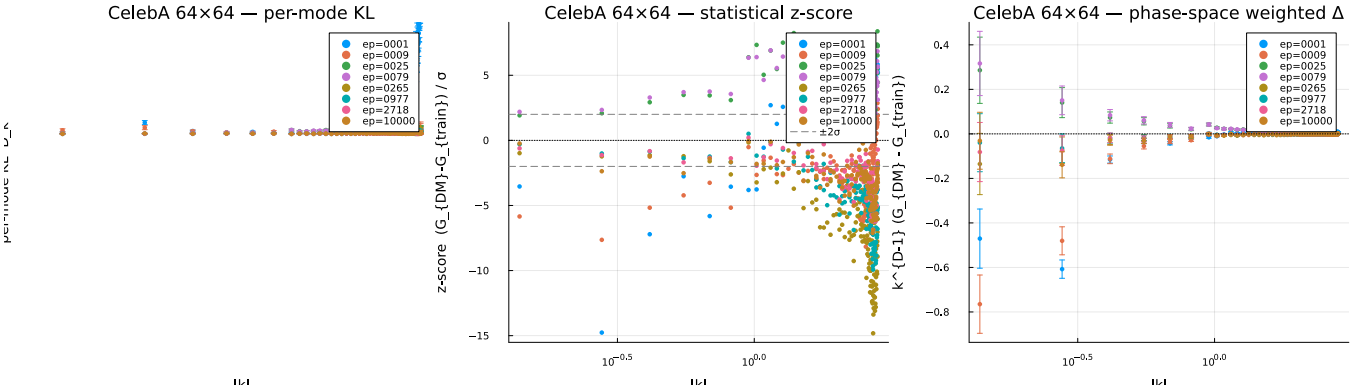

FIG. 19. Propagator-mismatch metrics for a NCSN++ diffusion model on CelebA grayscale 64×64 images at eight log-spaced training checkpoints. Left: per-mode Gaussian KL D_k defined in (25); the reference value $D_k = 0$ corresponds to exact propagator matching. Centre: statistical z-score of (26); the dashed lines mark the $\pm 2\sigma$ consistency band. Right: phase-space weighted propagator difference w_k of (27) with $D = 2$; integrating w_k over a momentum band returns the contribution of that band to the $\langle \phi^2 \rangle$ mismatch.

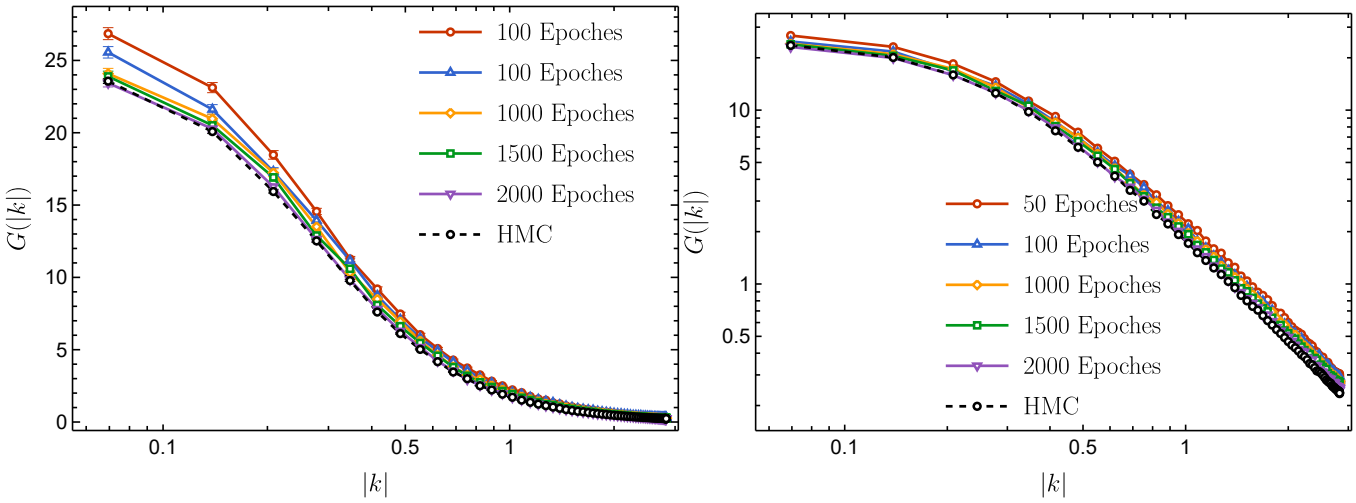

FIG. 20. Momentum-space propagator $G(p)$ for $\kappa = 0.26$ (symmetric phase), $L = 128$. Semi-log (left) and double-logarithmic (right) comparison of HMC and DM at different training epochs.

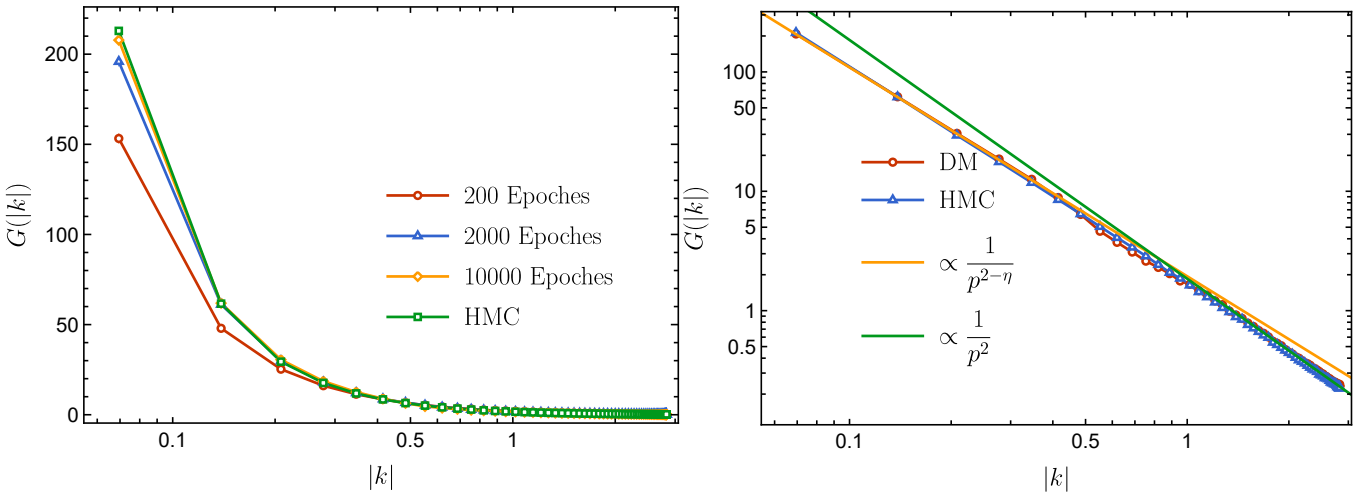

FIG. 21. Propagator $G(p)$ at $\kappa = 0.2705 \approx \kappa_c$ (near-critical), $L = 128$. Semi-log (left) and double-logarithmic (right) comparison of HMC and DM at different training epochs.

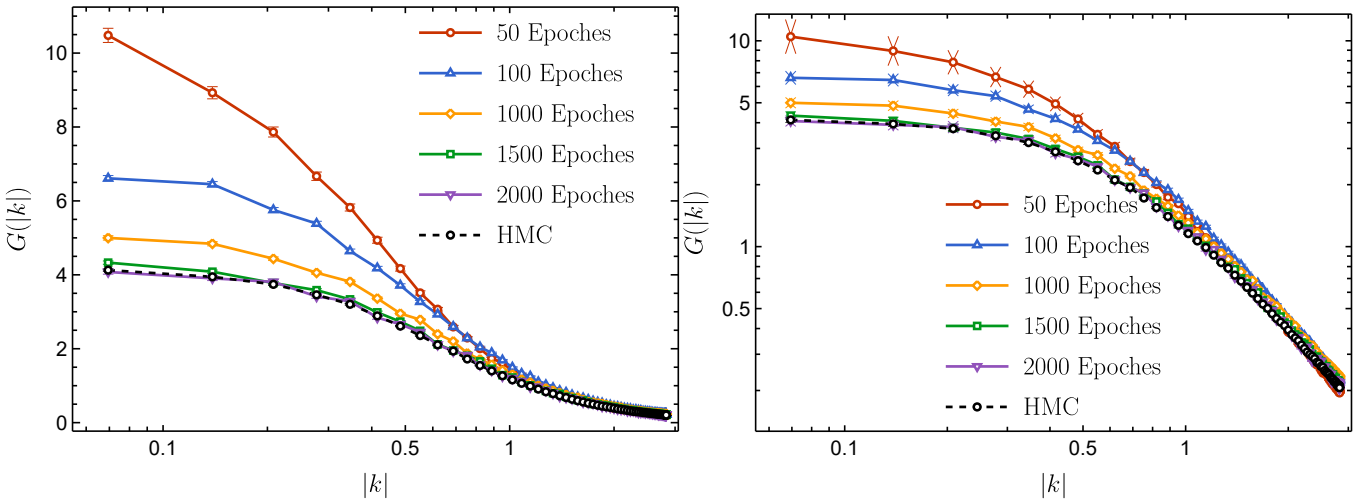

FIG. 22. Propagator $G(p)$ for $\kappa = 0.28$ (broken phase), $L = 128$. Semi-log (left) and double-logarithmic (right) comparison of HMC and DM at different training epochs.